

AI-ASSISTED PEER-REVIEW

So, how will GenAI change PhD research?

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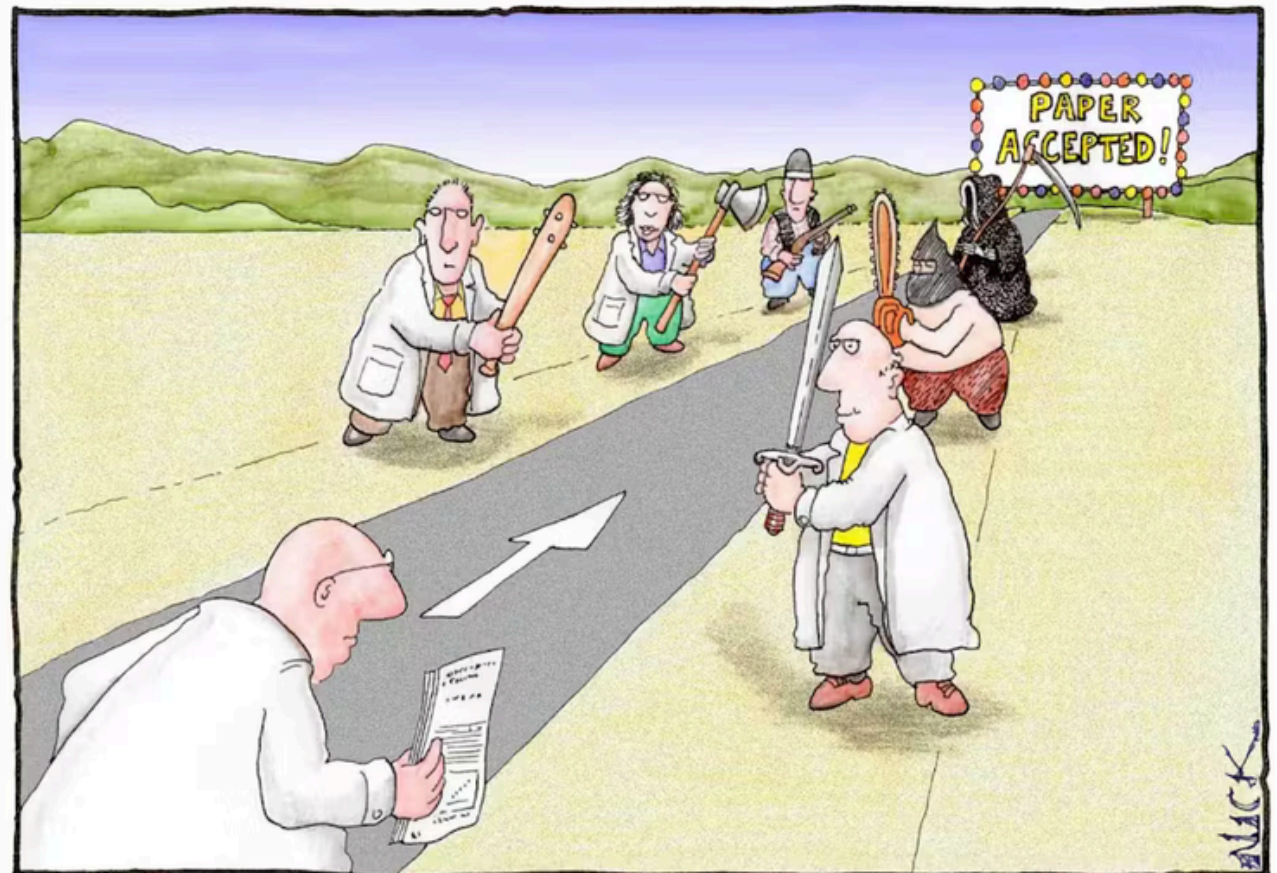
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PRESENT OF PEER REVIEW

BIAS

QUALITY
ISSUES

PREDATORY
JOURNALS

INCONSISTENCY

UNPAID OR
POORLY PAID

TIME
CONSUMING

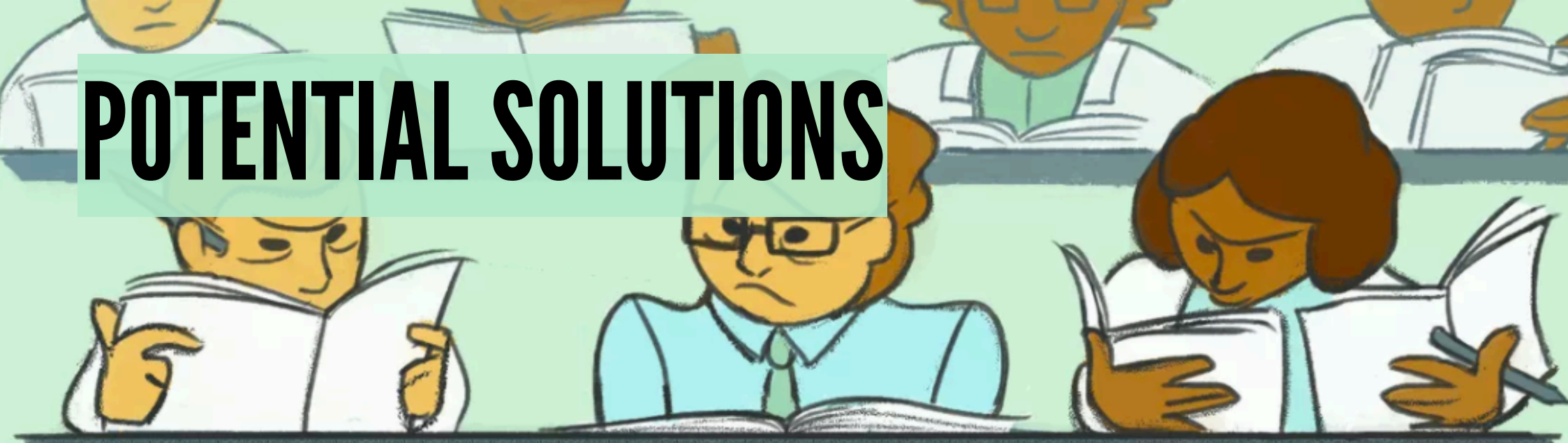
WASTED
COMMENTARY

ABUSE OF
PEER REVIEW



Early-career researchers are most affected.

POTENTIAL SOLUTIONS

An illustration of a classroom with several students sitting at desks, reading books. The students are depicted in a simple, cartoonish style with large heads and small bodies. The background is a light green color.

TRANSPARENT PEER REVIEW

INCREASING RELIABILITY

REVIEWER INCENTIVE

PRE-PRINT PEER REVIEW

IMPROVED RECRUITMENT

AI-ASSISTED PEER REVIEW



HOW GenAI FITS INTO PEER REVIEW? OR DOES IT?



IDENTIFY

- Plagiarism
- Incorrect formatting
- Language errors
- Scientific errors



MATCH WITH

Appropriate reviewers with **relevant** expertise to speed up peer-review process.



PREDICT

Likely feedback from peer reviewers based on the manuscript's content.

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AI, peer review and the human activity of science

When researchers cede their scientific judgement to machines, we lose something important.

By [Carl T. Bergstrom](#) & [Joe Bak-Coleman](#)

GenAI ASSISTANCE FOR STUDENTS



STRUCTURED FEEDBACK
GENERATION



AUTOMATED WRITING
FEEDBACK



DRAFTING REBUTTALS



FORMAT AND COMPLIANCE
CHECKS



GenAI ASSISTANCE FOR PROFESSORS

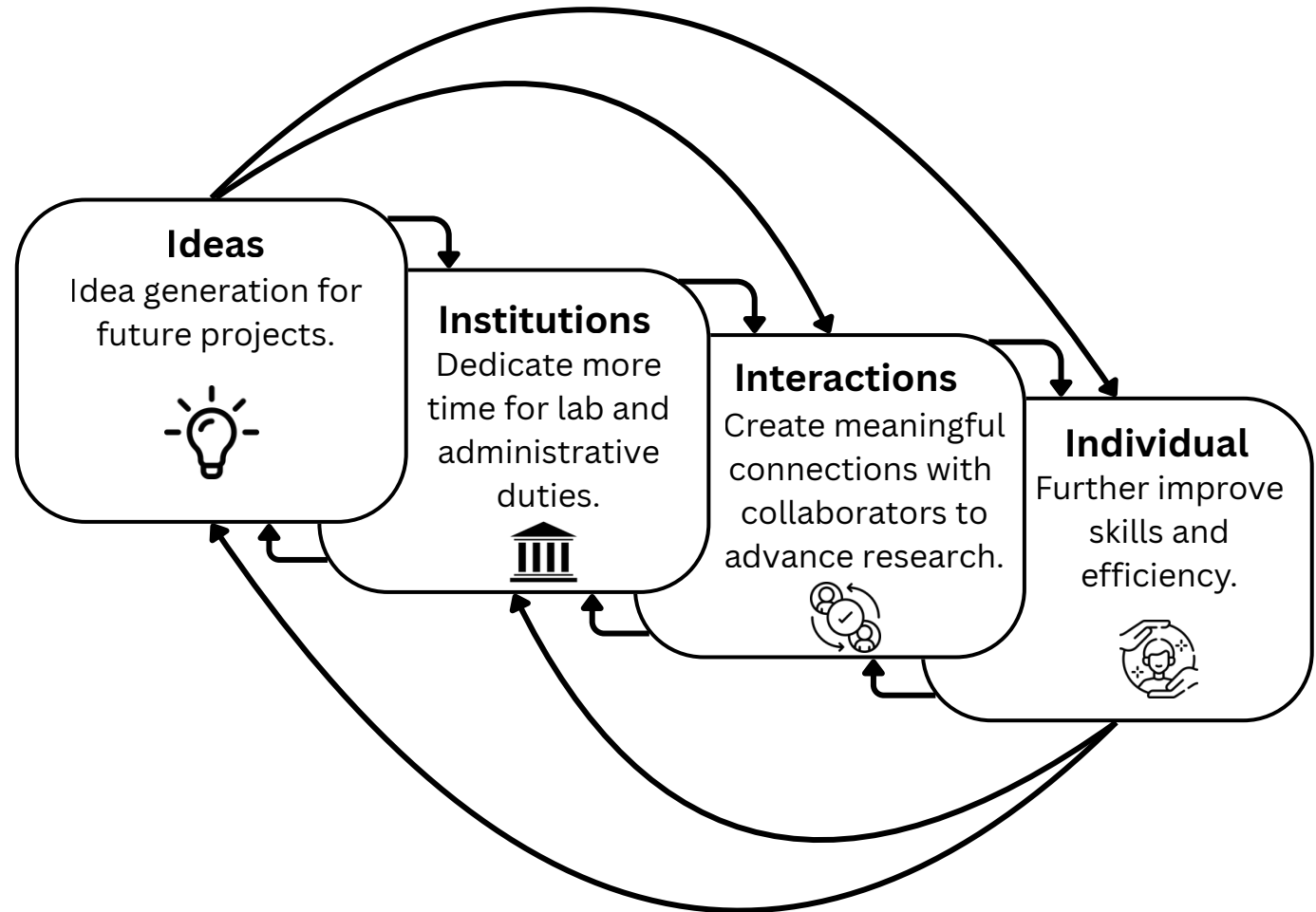
REDUCE ADMINISTRATIVE
BURDEN

ENHANCE
CONSISTENCY AND
FAIRNESS

DECISION MAKING
SUPPORT

INCREASE
TURNAROUND RATE

DIVERSIFY
PERSPECTIVES

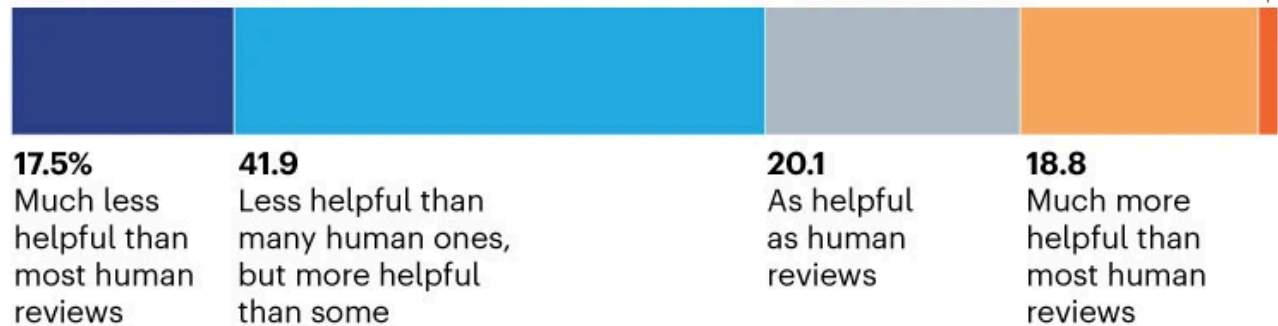


ETHICAL CONSIDERATIONS/LIMITATIONS

COMPARING AI AND HUMAN PEER REVIEW

A survey asked 300 researchers to compare human and LLM* reviews of their own papers. Respondents found LLM feedback slightly less helpful, on average, than human feedback, but LLMs performed better than some human reviewers.

LLM reviews were:



REGULATIONS ON AI USE IN PEER REVIEW

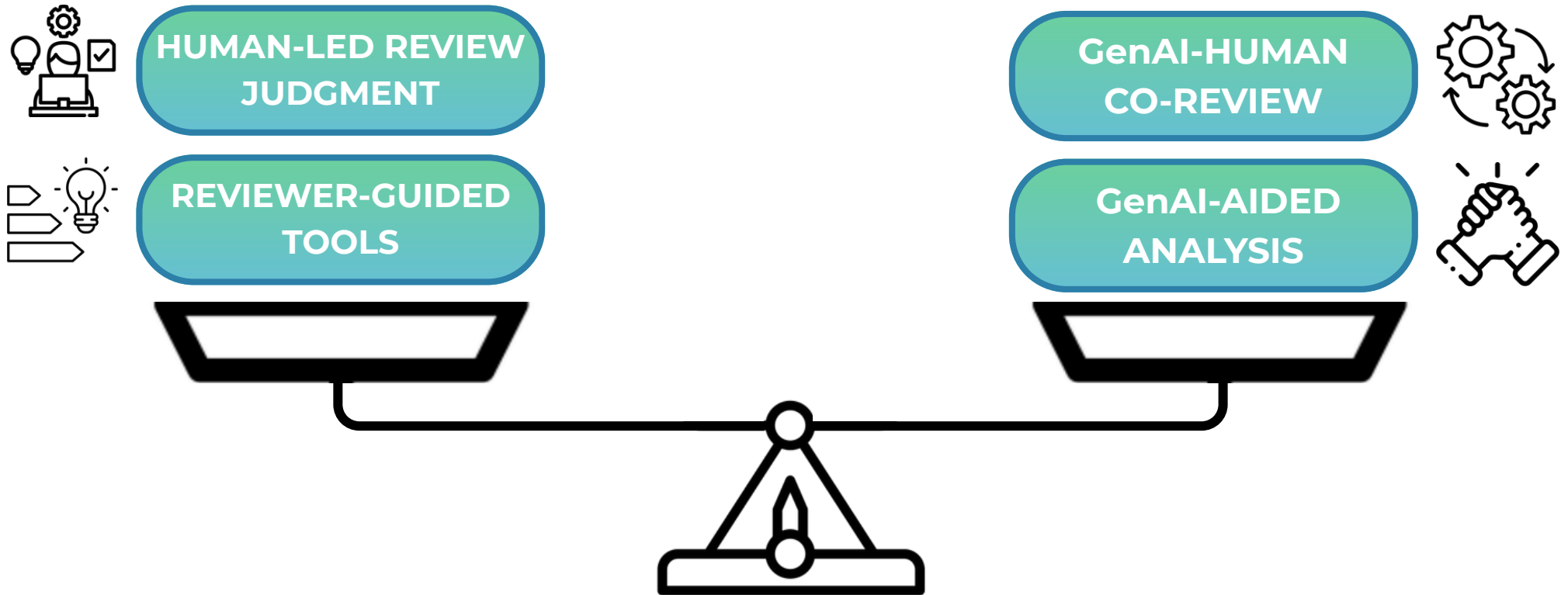
- Elsevier: bans
- Springer Nature: limited use
- Wiley: limited use
- ACS: bans
- Science: bans

59%

of the 78 top medical journals with guidance on the matter ban AI use in peer review.

Currently, GenAI is NOT a replacement, rather a thinking partner to assist researchers.

CONCLUSION



The future for PhD students lies in a balanced partnership.